UNIT 5 INTRODUCTION TO RENEWABLE ENERGY

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5.1 INTRODUCTION

Sources of energy are generally classified as renewable energy sources and non-renewable energy sources.

Renewable Energy Sources: The renewable energy or non-conventional energy source refer to sources which are almost unlimited or which can be replenished over a short span of time. Such sources are – sun, wind, water, agricultural residue, natural geysers, firewood, animal dung, etc. The total renewable energy sources in India have the potential to supply about 100,000 MW of power. Some of the major renewable energy sources are given in Figure 5.1.

Non-Renewable Energy Sources: The non-renewable energy sources are the fossil fuels such as coal, crude oil, natural gas and nuclear energy sources like uranium etc. (see Figure 5.2). The developed countries have about 20% of the world's population and use about 60% of the world's non-renewable energy resources.

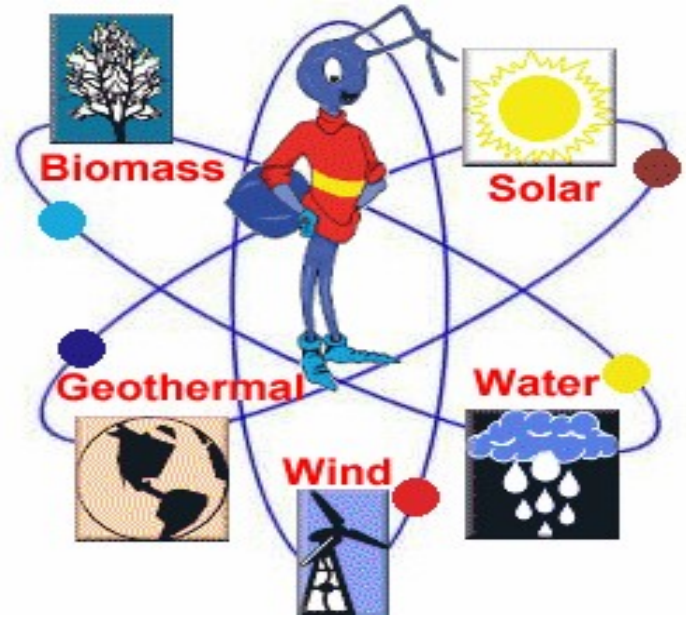


Figure 5.1: Renewable Energy Sources (*Source : eia.doe.gov*)

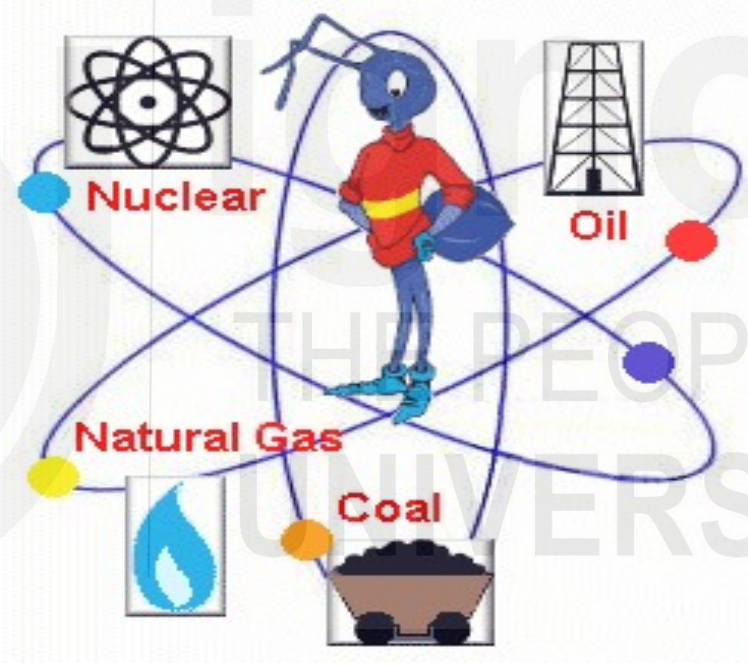


Figure 5.2: Non-renewable Energy Sources (*Source :* eia.doe.gov)

Let us briefly describe different forms of renewable energy.

Solar Energy: The energy generated from the sun is known as solar energy. India receives solar energy equivalent to over 5000 trillion KWh/year which is far more than total energy consumption in the country.

Hydel Energy: The energy generated from water is known as hydel energy.

Biomass Energy: The energy available from firewood and agro-residue is known as biomass energy. The traditional chulha, which is used in Indian villages, is an inefficient way of using biomass energy. Ninety per cent of the energy in the fuel is lost into the atmosphere; only ten per cent of the energy goes to actually heat the pot for cooking purposes.

Gasifiers convert wood, charcoal and other biomass to a combustible gas which may be used to produce electricity. About 1000 MW of power can be generated

from urban and municipal solid waste and up to 700 MW from Industrial waste in India. You may refer to Unit 2 for more details.

Biogas Energy: The energy available from animal dung and city biodegradable waste etc is called biogas energy. Depending on how much dung can be collected, biogas can meet the cooking energy needs of nearly 40 percent of the rural households of the country.

There are about 232 million cattles in the country. Even if one-third of the dung produced annually from these is available for biogas production and for recycling as farm manure, 12 million biogas plants of family size can be installed. Each biogas plant could save about 1260 Kg of fuel wood per year. You may refer to Unit 3 for more details

Wind Energy: The energy available from wind is known as wind energy. The Indian wind energy programme is one of the largest in the world, having an installed wind capacity of over 800 MW. India is ranked fifth in the world with a total wind power capacity of 1080 MW, out of which 1025 MW have been established in commercial projects. You may refer to Unit 4 for more details.

Geothermal Energy: The energy available from hot dry rocks, hot water springs, natural geysers, etc is known as geothermal energy.

All the renewable sources of energy are fairly non-polluting and considered clean.

You know that human society today need all kinds of technologies for their survival. Every day technology requires an energy source to work. In this process, we make use of many energy sources. It is the hard fact that waste products are produced when we use energy. The use of an energy source depends on the amount of energy available, ease of distribution and the extent to which energy can be concentrated and the convenience with which it can be used.

We only are responsible for the intelligent and safe use of energy and hence to save the environment.

Objective

After studying this unit, you should be able to

- know renewable energies in general and solar energy in particular and its potential
- understand various thermal applications of solar energy
- understand direct use of solar energy
- understand the environmental impacts of fossil fuels,
- understand the environmental impacts of nuclear energy,
- understand the environmental impacts of renewable energy, and
- understand the environmental impacts of green house gases.

5.2 EXPECTATIONS FROM THE RENEWABLE ENERGY DURING 11TH PLAN

As per the Rural Electrification Policy-REP (2006), the lifeline energy, i.e. minimum energy required by a house hold is 1 kWh/household/day. Renewable energy technologies can play a major role to achieve this target. During 11th plan the power generation from renewable sources, mainly small hydro, wind and bio is expected to be at least 12.5 per cent power generation installed capacity in the country.

The aim for the 11th Plan is a capacity addition of 15,000 MW from renewable. By the end of the 11th Plan, renewable power capacity could be 25,000 MW in a total capacity of 200000 MW accounting for 12.5 per cent and contributing around 5 per cent to the electricity mix.

Renewable power capacity by the end of the 13th plan period is likely to reach 54,000 MW, comprising 40,000 MW wind power, 6,500 MW small hydro power and 7,500 MW bio-power, which would correspond to a share of 5% in the then electricity-mix.

5.3 ESTIMATED POTENTIAL AND INSTALLED CAPACITY OF MAJOR RENEWABLE ENERGY TECHNOLOGIES IN INDIA

The estimated potential and current use of renewable energy is given in Table 1.1. 1 (MNRE, Govt. of India).

Table 1.1: Estimated Potential and Current Progress of Renewable Energy Use

S. No.	Source/system	Estimated potential	Achievement as on 31 January 2009
I	**Power from Renewables**		
A	Grid-interactive renewable power	(MW)	(MW)
1.	Wind power	45 195	9755.85
2.	Bio power (agro residues and plantations)	16 881	683.30
3	Bagasse cogeneration	5 000	1033.73
4	Small hydro power (up to 25 MW)	15 000	2344.67
5	Energy recovery from waste (MW)	2 700	58.91
6	Solar photovoltaic power	-	2.12
B	**Captive/combined heat and power/distributed renewable power**		**(MW)**
7	Biomass/cogeneration (non-bagasse)	-	150.92
8	Biomass gasifier	-	160.31
9	Energy recovery from waste	-	31.07
	Sub total (B)	-	342.30
	Total (A+B)	-	14220.22
II	**Remote village electrification**		**5410 villages/hamlets**
III	**Decentralized energy systems**		
10	Family –type biogas plants	120 lakh	40.90 lakh
11	Solar photovoltaic systems	50 MW/km^2	120 MWp
	i) Solar street lighting system	-	70 474 nos
	ii) Home lighting system	-	434692 nos
	iii) Solar lantern	-	697419 nos
	iv) Solar power plants	-	8.01 MWp
	v) Solar photovoltaic		7148 nos
12	Solar thermal systems		
	i) Solar water heating systems	140 million m^2 collector area	2.60 million m^2
	ii) solar cookers		6.37 lakh
13	Wind pumps		1347 nos
14	Aero generator/hybrid systems		0.89 MWeq
IV	**Awareness programmes**		
16	Energy parks	-	504 nos
17	Aditya Solar Shops	-	284 nos
21	Renewable energy clubs	-	521 nos
22	District Advisory Committees		560 nos

MW– megawatt; KW– kilowatt; MWp – megawatt peak; m^2 – square metre; Km^2 – kilometer square

Source: *MNRE, Govt. of India, Akshay Urja, Vol-2, issue 4th, Feb. 2009.*

The perspective plan for grid-interactive renewable power is summarized in Table 1.2.

Table 1.2: Perspective plan for grid-interactive renewable power for 2022, i.e. end of 13th Plan period (capacity in MW)

Resource	Upto 9th Plan	10th Plan	11th Plan	12th and 13th Plans	Total
Wind power	1667	5333	10500	22,500	40000
Small hydro power	1438	522	1400	3140	6500
Bio power	368	669	2100	4363	7500
Solar Power	2	1	-	-	3
Total	**3475**	**6525**	**14000**	**30003**	**54003**

SAQ 1

(a) State the norm of minimum lifeline supply of energy

(b) State the expectations from the renewable energy during 11th Plan

5.4 ADVANTAGES AND DISADVANTAGES OF RENEWABLE ENERGY

The renewable energy or alternative energy sources are solar, wind, biomass, hydro, geothermal, ocean thermal, tidal etc. All these renewable energies are having several advantages as well as disadvantages. A bird eye view is given in Table 1.3. A quick view of the advantages of renewable energy is given in Figure 5.3.

Table 1.3: Advantages and Disadvantages of Renewable Energy

Renewable Energy	Advantages	Disadvantages
Solar	Always available as long as sun is shining No pollution	Low efficiency High initial costs Lack of storage
Wind	Power in windy areas No pollution	Highly variable source Low efficiency (about 30%) Efficient energy storage needed
Hydro	High Efficiency No pollution Little waste heat Low cost to the user	May alter hydrological cycle Change watershed characteristics
Geothermal	High efficiency Low initial costs	Highly local resource Non-renewable
Biomass	Biomass are natural Re-use is attractive Gives cogeneration facilities Practical for individual farmer	Pollution from biomass burners Transport is difficult because of moisture
Tidal	Steady source Capable of exploiting tides for maximum efficiency	Low duty cycle Huge modification in coastal environment High costs
Ocean Thermal	Enormous energy Steady flow Large scale use is possible	Highly technical Damage to coastal environment

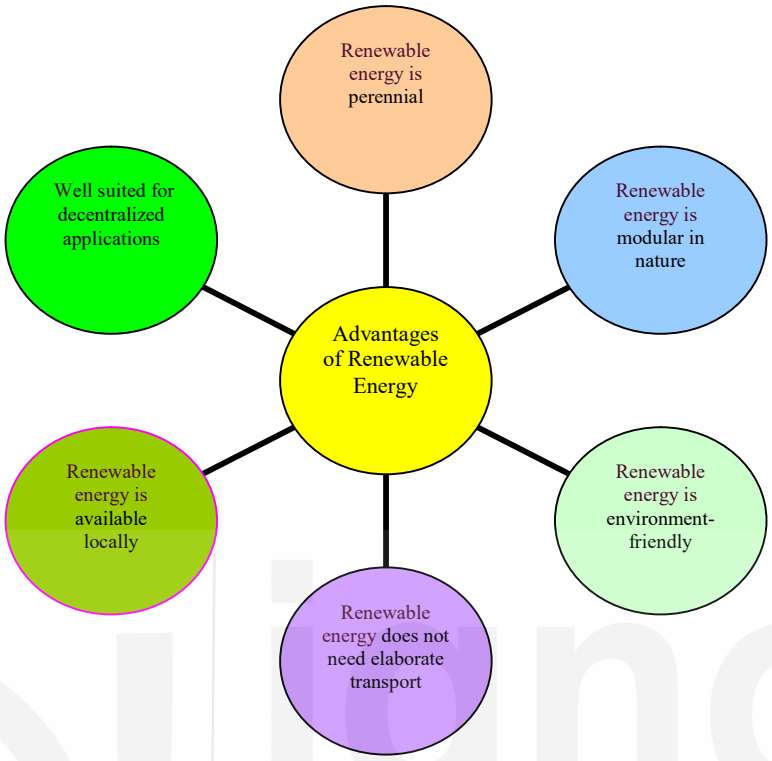


Figure 4.3: Advantages of Renewable Energy

We will now discuss solar energy. The other forms of renewable energies will be discussed in the subsequent units.

5.5 SOLAR ENERGY POTENTIAL

You must have learnt in your school text books that Sun is a fusion reactor and is the ultimate source of energy. You may be aware that Sun is located at a distance of 150 million kilometer from the Earth and is about hundred times bigger than the earth in size. Most of the earth's energy comes from the sun. Our sun and the other stars are nuclear reactors that fuse hydrogen atoms together to form helium atoms. A large amount of energy is released in the process. The reaction is similar to what goes on in the explosion of a hydrogen bomb.

In fact all forms of energy that human beings have consumed so far and are still consuming are in one way or another due to sun.

Point to Remember

Solar energy comes to us in the form of electromagnetic radiation. Sun is the ultimate source of energy. Some interesting facts about Sun are the following:

- A sun ray emitted from the Sun, travelling at the speed of 3×10^8 km/s, takes about 9 minutes to reach the Earth.
- The earth receives about 1.3×10^{17} W/h. of energy.
- The temperature of the Sun at the centre is 15 million ^{0}K and at the surface it is about 6000°K.

About 50% of the energy received outside Earth's atmosphere actually reach the Earth and is about 698 joules/sq. m.sec at sea-level.

Point to Remember

We receive every minute enough energy on the earth to meet our demand if harnessed properly.

Solar energy is the most readily available source of energy. It does not belong to anybody and is, therefore, free. It is also the most important of the non-conventional sources of energy because it is non-polluting and, therefore, helps in reducing the greenhouse effect.

Solar energy has been used since ancient times, but in a most primitive manner. Before 1970, some research and development was carried out in a few countries to exploit solar energy more efficiently, but most of this work remained mainly academic. After the dramatic rise in oil prices in the 1970s, several countries began to formulate extensive research and development programmes to exploit solar energy.

When we hang out our clothes to dry in the sun, we use the energy of the sun. In the same way, solar panels absorb the energy of the sun to provide heat for cooking and for heating water. Such systems are available in the market and are being used in homes and factories.

In the next few years it is expected that millions of households in the world will be using solar energy. India is one of the few countries with long days and plenty of sunshine. Solar energy could be easily harnessed. Solar thermal energy is being used in India for heating water for both industrial and domestic purposes.

5.6 SOLAR ENERGY USES

India receives solar energy equivalent to over 5000 trillion kWh/year, which is far more than the total energy consumption of the country. In principle solar energy can also be used to meet all our energy needs, both thermal as well as electricity. The solar energy can be broadly classified in two categories on the basis of its use as shown in Figure 5.4:

(1) Solar Active Applications i.e. direct use of solar energy to produce electricity

(2) Solar Passive Applications i.e. indirect use of solar energy usually called solar thermal applications.

Thermal energy can be used for Cooking/Heating, Drying/Timber seasoning, Distillation, Electricity/Power generation, Cooling, Refrigeration, Cold storage, etc.

Some of the gadgets and other devices for exploiting thermal energy are :

Solar cooker, flat plate solar cookers, concentrating collectors, solar hot water systems (domestic and industrial), solar pond, solar hot air systems, solar dryers, solar timber kilns, solar stills, solar pond, concentrating collectors, air conditioning, solar collectors coupled to absorption and refrigeration systems, etc. Although solar photovoltaic systems are used to convert solar energy directly into electricity, the heat generated during the process can be used for solar thermal applications.

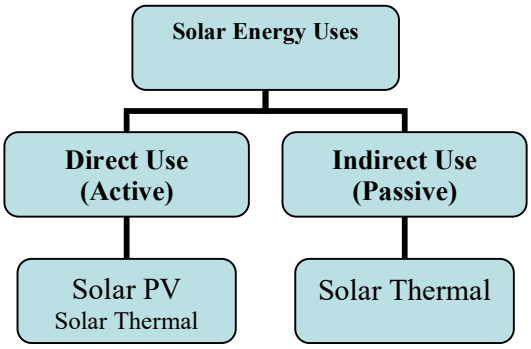


Figure 5.4: Solar Energy Uses

The solar energy is directly converted in the desired application form. It is usually divided into following two forms :

(1) Solar thermal for heating applications

(2) Solar Photovoltaic for electricity generation

The use of solar energy for thermal applications is called Solar Thermal Applications. Here solar energy is collected and then converted to heat energy for applications such as water and air heating, cooking and drying, steam generation, distillation by making use of an appropriate solar energy device.

Basically a solar thermal device consists of the following :

(a) a solar energy collector called absorber

(b) a heat transferring medium

(c) a heat storage or heat tank

You will now learn the basic concepts of these components.

5.7 DIRECT USE OF SOLAR ENERGY

The solar energy is directly converted in the desired application form by using Solar Cell technology. You will earn more about this in OEY 002, block 2. Solar photovoltaic technology is now gaining importance because we are running out fossil fuel in the times to come.

Through Solar Photovoltaic (SPV) cells, solar radiation gets converted into DC electricity directly. This electricity can either be used as it is or can be stored in the battery. This stored electrical energy then can be used at night. SPV can be used for a number of applications such as: domestic lighting

- street lighting
- village electrification
- water pumping
- railway signals.

If all the means to make efficient use of solar energy are implemented, it would reduce our dependence on non-renewable sources of energy and make our environment cleaner.

SAQ 2

Identify which one is true: Renewable energy is

(a) perennial

(b) available locally

(c) does not need elaborate arrangements for transport

(d) modular in nature

(e) environment-friendly

(f) well suited for decentralized applications

(g) well suited for use in remote areas.

(h) All above

5.8 MODEL SOLAR CITIES

The MNRE (Ministry of New and Renewable Energy), Govt. of India proposes to develop 60 solar cities during the Eleventh Plan Period. At least one city in each state, to a maximum of five cities in a state, will be supported by the ministry. This scheme has been developed to meet the peak electricity demand of cities, reduce dependence on fossil fuels and expensive oil and gas for energy, and promote increased use of renewable energy.

MNRE, Govt. of India has already launched a programme of developing Nagpur as a model solar city in 2009. Nagpur will be the first solar city in the country and will become a model solar city by 2012. Up to 10% of energy consumption of this city has been targeted to be met through renewable energy and energy efficiency measures. Major solar energy systems will also be installed including streetlights, garden lights, traffic lights, hoardings, and solar water heaters. Energy-efficient green buildings will also be promoted on a large scale in the city.

5.9 ADVANTAGES AND DISADVANTAGES OF SOLAR ENERGY

Solar energy is available in the most natural and attractive form. It has wide-spread distribution, it is time-specific and definite, it has virtually inexhaustible supply, it has antibacterial and disinfecting qualities, and with proper exposure to human body, it can provide Vitamin-D, helpful for building of bones. It is environment-friendly and has now emerged as an effective alternative source of power. Solar power is supposed to be better than nuclear power. We will now discuss advantages and disadvantages of solar energy.

Advantages of solar energy:

- Solar energy is free.
- Solar energy does not produce waste and pollution.
- In sunny areas having no grid power, solar power can be used.
- Low power areas are very handy like solar powered batteries, lights, etc.

Solar power is reliable. Sun will continue to shine so there is a sense to use solar energy.

Disadvantages of solar energy

- Solar energy is not available in the night.
- Solar power stations are relatively expensive.

5.10 UNDERSTANDING ENERGY, ECONOMY AND ENVIRONMENT

When we try to understand energy, we have to understand energy resources and their limitations along with their impact on environment. All the developed nations of the world are consuming more energy for their economic development as compared to developing nations. As a result of this, they are releasing more pollutants to the environment. Thus, we can say that Energy, Economy and Environment are closely related. For the all round development of human society, we need energy which affects the environment. It is also interesting to see that a major portion of our energy consumption is derived from fossil fuels (about 90%). With this level of fossil fuels consumption, we are increasing the emissions of CO_2 and other gaseous pollutants.

We can help to save the environment by :

(1) Reducing the use of energy without affecting the progress,

(2) Reducing the waste produced, and

(3) Using renewable energy sources.

5.11 ENERGY AND MATTER

Let us elaborate the concept that all changes in the universe involve interactions between matter and energy. This is shown in Figure 5.5.

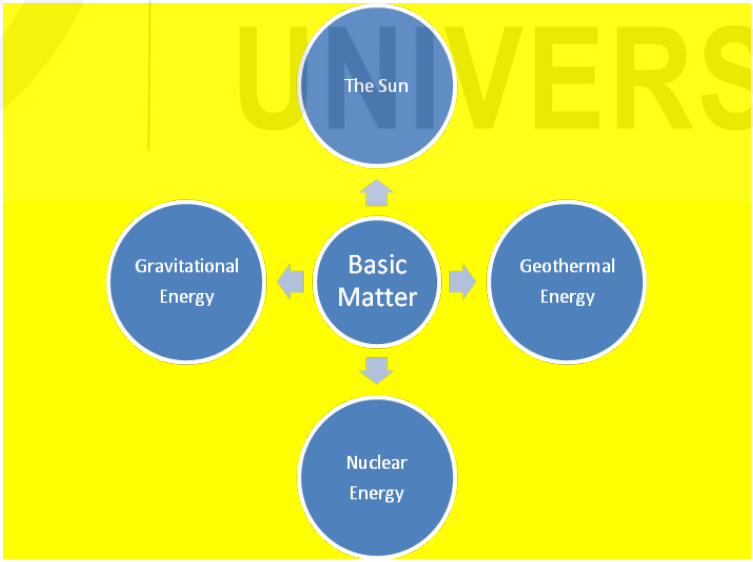


Figure 5.5: Interaction between Matter End Energy

The solar energy, gravitational energy, geothermal energy and nuclear energy when converted to useful energy and used also result into waste heat losses to the environment.

Point to Remember :

Global warming and climate change, primarily due to the emission of greenhouse gases like CO_2 (carbon dioxide), are the two major concerns facing the world today.

If only 3% Indians were to cook with solar cookers, we would save 3.2 MT (million tonnes) of wood per year and reduce CO_2 emissions by 6.7 MT per year.

5.12 ENVIRONMENTAL IMPACTS OF FOSSIL FUELS

5.12.1 Human Activities Responsible for Greenhouse Gases

The important activities responsible for releasing greenhouse gases into the atmosphere are:

(1) Production of carbon dioxide when fossil fuels are used to generate energy and when forests are cut down and burned.

(2) Production of methane and nitrous oxide from agricultural activities, changes in land use, and other sources.

(3) Production of man-made chemicals called halocarbons (CFCs, HFCs, PFCs) and other long-lived gases such as sulphur hexafluoride (SF6) by industrial processes.

The observations of earth temperatures revels that over the last 100 years, the earth is getting warmer and warmer, as compared to previous 8000 years when temperatures have been relatively constant. The present temperature is 0.3-0.6ºC warmer than it was 100 years ago.

Let us briefly discuss the damage to the ozone layer caused by greenhouse gases. The hole in the upper ozone layer (stratosphere) causes cooling, because ozone itself is a greenhouse gas. However, ozone is needed in the stratosphere, where it occurs naturally to filter out harmful ultraviolet radiation.

Ozone in the lower atmosphere (troposphere) is caused by pollutants. Ozone in the troposphere is a danger to human health, because it can cause lung damage and other cardio-pulmonary problems when inhaled.

You will be surprised to know that carbon dioxide emissions weigh more than the original fuel. The reason is as follows:

In combustion process of fossil fuels, each carbon atom in the fuel combines with two oxygen atoms in the air to make carbon dioxide ($C + O_2 = CO_2$). The addition of two oxygen atoms to each carbon atom forms carbon dioxide, which has an atomic weight of 44 – roughly 3.67 times more than the atomic weight of the carbon (12).

Example 5.1

Sub-bituminous coal has 51% carbon. The carbon dioxide emissions are about 3.67 times the weight of the coal's carbon. Determine by what percentage this emission is more than the original coal.

Solution

Let us assume that weight of coal is 100 Kg.

Weight of Carbon = 51% = 51 Kg.

Weight of CO_2 = 3.67 × weight of Carbon = 3.67 × 51 = 187 Kg.

Weight of excess CO_2 = 187 – 100 = 87 Kg

i.e. 87%.

5.12.2 Options for Limiting Emissions

The options available for limiting emissions to the environment are :

(1) Encouraging energy efficiency in industries, homes, offices, vehicles, and agriculture.

(2) Encouraging use of energy efficient technologies.

(3) Good housekeeping measures.

(4) Ensuring the interaction between energy, economy and environment

(5) Encouraging energy substitution which means use of renewable energy technologies as much as possible.

(6) Reducing the usage of electricity, LPG, petrol, diesel etc wherever possible.

5.12.3 Global Climate Change Treaty : The Kyoto Protocol

The Kyoto Protocol expects that the industrialized nations should make efforts in reducing emissions of greenhouse gases. The six major greenhouse gases covered by the protocol are Carbon dioxide (CO_2), Methane (CH_4), Nitrous oxide (N_2O), Hydrofluorocarbons (HFCs), Perfluorocarbons (PFCs), and sulfur hexafluoride (SF_6).

The United States would be obligated under the Protocol to a cumulative reduction in its greenhouse gas emissions by 7% below 1990 levels for three greenhouse gases (including carbon dioxide), and below 1995 levels for the three man-made gases, averaged over the commitment period 2008 to 2012.

The Protocol states that "developed countries are committed, individually or jointly, to ensuring that their aggregate anthropogenic carbon dioxide equivalent emissions of greenhouse gases do not exceed amounts assigned to each country".

India has ratified the contentious Kyoto Protocol in August 2002. However, the Kyoto Protocol does not set any binding limits on developing nation emissions, nor does it establish a mechanism or timetable for these countries to take on such limits voluntarily. On the other hand, the Protocol does establish a so-called Clean Development Mechanism (CDM), which allows developed countries to invest in projects in developing countries that reduce greenhouse gas emissions and receive credit for the reductions.

5.12.4 India's Greenhouse Gas Emissions

India has experienced a dramatic growth in fossil fuel CO_2 emissions, and the data compiled by various agencies shows an increase of nearly 5.9 % since 1950. At present India is rated as the 6th largest contributor of CO_2 emissions behind China, the 2nd largest contributor. However, our per capita CO_2 emissions of 0.93 tonnes per annum are well below the world average of 3.87 tonnes per annum. Fossil fuel emissions in India continue to result largely from coal burning.

5.12.5 Clean Development Mechanism (CDM)

The Clean Development Mechanism (CDM enables developing countries to participate in joint greenhouse gas (GHG) mitigation projects. Under this Protocol, countries (developed countries and economies in transition) are required

to reduce GHG emissions to below their 1990 levels. The CDM enables these countries to meet their reduction commitments in a flexible and cost-effective manner. It allows public or private sector entities to invest in GHG mitigation projects in developing countries. In return the investing parties receive credits or Certified Emission Reductions (CERs), which they can use to meet their target.

The ultimate source of fossil fuels is the sun as shown in Figure 5.6. The coal is the result of plant life. The coal gets converted into mechanical energy resulting into waste heat loss to the environment. The oil and gas results through marine life and may be converted into mechanical energy resulting into waste heat loss to the environment.

All these power generation by using fossil fuels must include consideration of hazards to the environment. The energy conversion and use affect plant and animal life, agricultural productivity, air and water purity, landscape beauty, and climate. Human hazards due to power generation include occupational dangers in mining, transporting, and processing of fuels. They also include public health hazards associated with power plant emissions, particularly in case of accidents. The main concern is air pollution due to the burning of fossil fuels.

The effects of fossil fuels pollution on the environment are mainly to the air and water and to a larger extent the land. The most important contaminants are oxides of sulphur (i.e. SO_2, hydrogen sulphide H_2S, and various sulphates), oxides of nitrogen (nitric oxide NO, nitrous oxide N_2O).



Figure 5.6: Environmental Impacts of Fossil Fuels

5.13 ENVIRONMENTAL IMPACTS OF NUCLEAR ENERGY

The ultimate source of nuclear energy is also the sun as shown in Figure 5.7. The nuclear energy gets converted into mechanical energy and electricity resulting into waste heat loss to the environment.

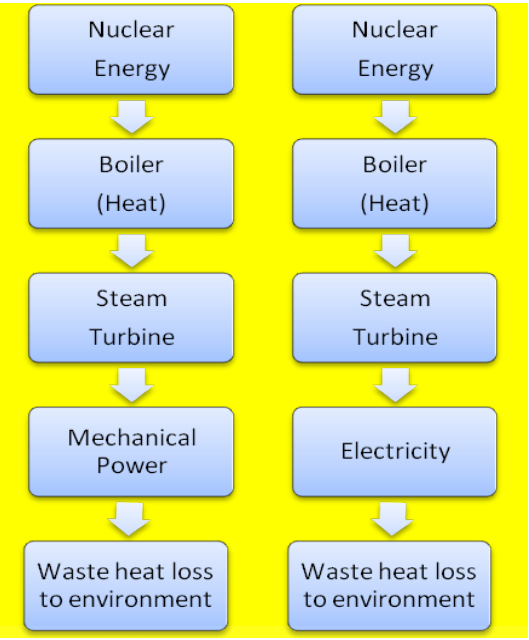


Figure 5.7: Environmental Impacts of Nuclear Energy

5.14 ENVIRONMENTAL IMPACT OF RENEWABLE ENERGY

The use of renewable energy does not add significant amount of waste heat to the environment. However, it must be understood that the energy used in the manufacturing of renewable energy technologies does add the waste heat loss to the environment.

You now know that photovoltaic energy constitutes an inexhaustible source of energy, and is less damaging for the environment. The effects of photovoltaic energy upon the main environmental factors are the following :

(a) The contribution to the greenhouse effect is inexistent. This is because generation of electrical energy directly from solar light does not require any kind of combustion, neither thermal pollution nor CO_2 emissions are produced.

(b) Solar PV cells are manufactured from silicon which can be obtained from sand.

The solar PV energy represents one of the best solutions for those places where there is a need for electrical energy whilst preserving the environmental conditions. The various conversion routes of renewable energy are shown in Figures 5.8 and 5.9.

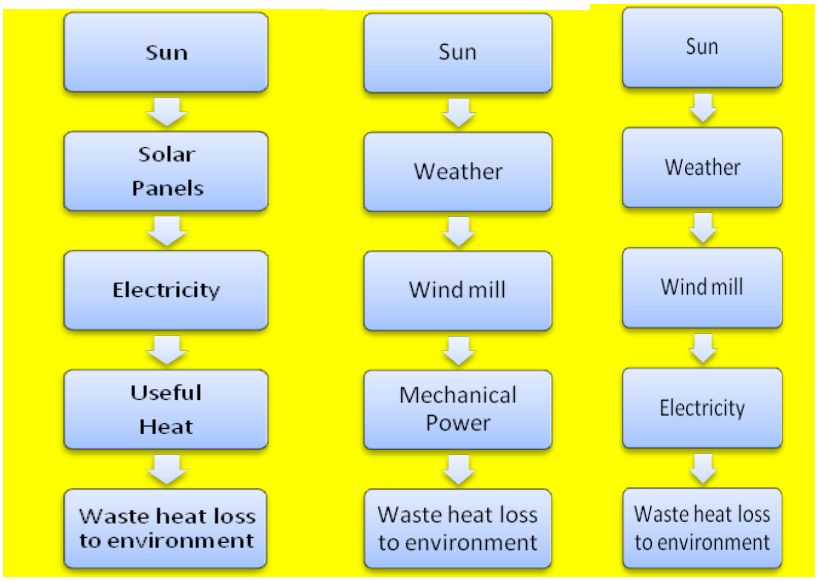


Figure 5.8 : Environmental Impacts of Renewable Energy

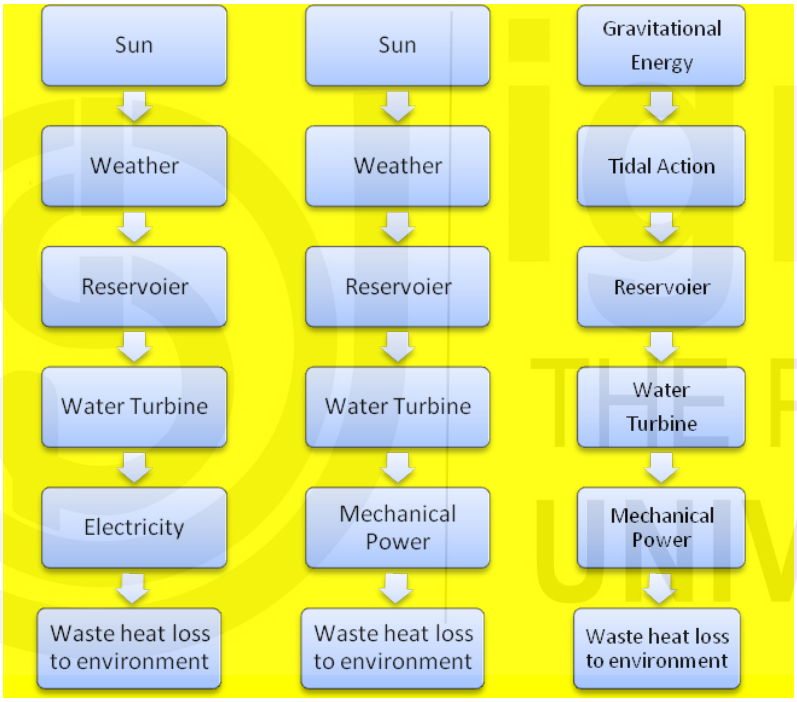


Figure 5.9: Environmental Impacts of Renewable Energy

SAQ 3

What are the environmental impacts of photovoltaic solar energy?

5.15 GREENHOUSE GASES

The main greenhouse gases are carbon dioxide, methane, nitrous oxide, and fluorocarbons. The greenhouse gases naturally blanket the Earth and keep it about 33 degrees Celsius warmer than it would be without these gases in the atmosphere.

The "greenhouse effect" is usually defined as the heating of the Earth's atmosphere due to the presence of greenhouse gases. The temperature of the Earth

has reportedly increased by about 0.5ºC during the past century; a phenomenon referred to as Global Warming. After the 1997 Kyoto Protocol, the world has finally taken the first step in reducing emissions.

The concept of green house effect may be understood by observing the following steps:

- Shorter-wavelength solar radiation from the sun passes through Earth's atmosphere.
- These radiations are absorbed by the surface of the Earth, causing it to warm.
- Part of the absorbed energy is then reradiated back to the atmosphere as long wave infrared radiation.
- Some of these long wave radiations escape back into space; the radiation cannot pass through the greenhouse gases in the atmosphere. Greenhouse gases like carbon dioxide, water vapor, and methane help inhibit the escape of radiant heat back into space.
- The greenhouse gases selectively transmit the infrared waves, trapping some and allowing some to pass through into space. The greenhouse gases absorb these waves and reemits the waves downward, causing the lower atmosphere to warm.

A brief description of the greenhouse gases is given below:

Carbon Dioxide

Carbon Dioxide (CO_2) is a colorless, odorless non-flammable gas and is the most prominent greenhouse gas in Earth's atmosphere. You may be aware that CO_2 is recycled through the atmosphere by the process called photosynthesis, which makes human life possible. **Photosynthesis is the process used by green plants and other organisms transforming light solar energy into chemical energy.** The plants absorb solar energy and convert CO_2, water, and other minerals into oxygen and energy rich organic compounds.

CO_2 is emitted into the air by various activities. The burning of fossil fuels and deforestation are the main activities. Every year humans add over 30 billion tons of CO_2 in the atmosphere.

Methane

Methane is a colorless, odorless and flammable gas. It is formed when plants decay and where there is very little air. Since 1750, methane has doubled, and could double again by 2050. Each year we add 350-500 million tons of methane to the air by raising livestock, coal mining, drilling for oil and natural gas, rice cultivation, and garbage sitting in landfills.

(www.envirolink.org/orgs/edf/sitemap.html).

N**itrous Oxide**

Nitrous oxide is another colorless greenhouse gas with sweet odor. It is primarily used as an anesthetic because it reduces pain and for this characteristic is called “laughing gas”. This gas is released naturally from oceans and by bacteria in soils. Nitrous oxide gas has risen by more than 15% since 1750.

Fluorocarbons

Fluorocarbons are general term for any group of synthetic organic compounds that contain fluorine and carbon. The chlorofluorocarbons (CFCs), can be easily converted from gas to liquid or liquid to gas. CFCs are widely used in refrigerators and air conditioners. When CFCs are emitted into the atmosphere, they break down ozone molecules in the Earth's ozone layer. The substitute for CFCs is hydro fluorocarbons (HFC's). HFCs do not harm or breakdown the ozone molecule, but they do trap heat in the atmosphere, making it a greenhouse gas, aiding in global warming. HFCs are used in air conditioners and refrigerators.

5.16 LET US SUM UP

Energy, Economy and Environment are closely related. For the all round development of human society, we need energy which affects the environment. It is also interesting to see that a major portion of our energy consumption is derived from fossil fuels (about 90%). With this level of fossil fuels consumption, we are increasing the emissions of CO_2.

We can help to save the environment by :

- Reducing the use of energy without affecting the progress,
- Reducing the waste produced, and
- Using renewable energy sources.

The "greenhouse effect" is usually defined as the heating of the Earth due to the presence of greenhouse gases. The temperature of the Earth has reportedly increased by about 0.5ºC during the past century; this is widely referred as Global Warming.

Human society today needs all kinds of technologies for their survival. Every day technology requires an energy source to work. In this process, we make use of many energy sources. It is the hard fact that waste products are produced when we use energy. The use of an energy source depends on the amount of energy available, ease of distribution and the extent to which energy can be concentrated and the convenience with which it can be used.

The sun, gravitational energy, geothermal energy and nuclear energy when converted to useful energy and used also results into waste heat losses to the environment. Let us consider various interactions and conversion processes involved.

Carbon Dioxide – A heavy colorless gas that does not support combustion, dissolves in water to form carbonic acid, is formed especially combustion of fossil fuels, is absorbed from the air by plants in photosynthesis, and is used in the carbonation of beverages.

Methane – A colorless, odorless, flammable gaseous hydrocarbon that is a product of decompositions of organic matter and of the carbonization of coal, is used as a fuel and as a starting material in chemical synthesis.

Photosynthesis – Synthesis of chemical compounds with the aid of solar energy formation of carbohydrates from carbon dioxide and a source of hydrogen in the chlorophyll containing tissues of plants exposed to light.

India lies in the sunny regions of the world. India receives solar energy equivalent to over 5000 trillion KWh/year. This is far more than the total energy consumption in the country. Daily average of incident solar energy ranges from 4 to 7 KWh/m^2 depending upon the location. The highest annual radiation energy

is received in western Rajasthan while the north-eastern region of the country receives the lowest annual radiation. It has been estimated that we have 250-300 days of clear sunny weather. Annual radiation being 1600 to 2200 KWh/m^2, even if 1 per cent of the nation's land is used, we could have nearly 1000 giga watts (GW) of power.

Solar energy, experienced by us as heat and light, can be used through two routes: the thermal route uses the heat for water heating, cooking, drying, water purification, power generation, and other applications; the photovoltaic route converts the light in solar energy into electricity, which can then be used for a number of purposes such as lighting, pumping, communications, and power supply in un-electrified areas.

Energy from the sun has many features, which make it an attractive and sustainable option: global distribution, pollution free nature, and the virtually inexhaustible supply.

5.17 KEY WORDS

Biomass

A renewable source of energy derived from organic matter like wood, agriculture waste; also include algae, sewage and other organic substances that may be used to make energy through chemical processes

Bio-fuels

Fuels made from biomass; include ethanol, biodiesel and methanol

Biogas

A combustible gas derived from decomposing biological waste; normally consists of 50 to 60 percent methane

Geothermal Energy

The energy available from hot dry rocks, hot water springs, natural geysers, etc is known as geothermal energy.

Hydel Energy

The energy generated from water is known as hydel energy.

Non-renewable Energy

The nonrenewable sources are the fossil fuels such as coal, crude oil and natural gas

Renewable Energy

Energy resources that constantly renew themselves or that are regarded as practically inexhaustible; include solar, wind, geothermal, hydro and wood

Renewable Resources

Renewable energy resources are virtually inexhaustible in duration; include biomass, hydro, geothermal, solar, wind, ocean thermal, wave, and tidal.

Solar Energy

The energy available from the sun is called solar energy. Solar energy can also be used to meet our energy needs, both thermal as well as electricity requirements.

Wind Energy

The energy available from wind is known as wind energy.

Carbon Dioxide

A heavy colorless gas that does not support combustion. It is released during combustion of fossil fuels and is absorbed from the air by plants in photosynthesis.

Methane

A colorless odorless flammable gaseous hydrocarbon that is a product of decompositions of organic matter and of the carbonization of coal.

Photosynthesis

Synthesis of chemical compounds with the aid of solar radiations, water and chlorophyll by the plants.

Greenhouse Gases

The main greenhouse gases are carbon dioxide, methane, nitrous oxide, and fluorocarbons. These gases are responsible for the heating of the earth's atmosphere.

5.18 ANSWERS TO SAQs

SAQ 1

(a) The lifeline energy norm of is 1kWh/household/day, as given in the Rural Electrification Policy -REP (2006).

(b) Power generation from renewable sources, mainly small hydro, wind and bio is expected to be at least 10 per cent power generation installed capacity in the country. Out of the overall target of 70,000 MW power generation installed capacity addition during the 11th Plan period, 14,500 MW (about 20%) capacity additions is proposed from renewable.

SAQ 2

(a)

SAQ 3

Photovoltaic energy constitutes an inexhaustible source of energy, and is less damaging for the environment. The effects of photovoltaic energy upon the main environmental factors are the following:

The contribution to the greenhouse effect is inexistent. This is because generation of electrical energy directly from solar light does not require any kind of combustion, neither thermal pollution nor CO_2 emissions are produced. Solar PV cells are manufactured from silicon which can be obtained from sand.

The solar PV energy represents one of the best solutions for those places where there is a need for electrical energy whilst preserving the environmental conditions.

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Web Link for more information on Energy

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